

Introduction

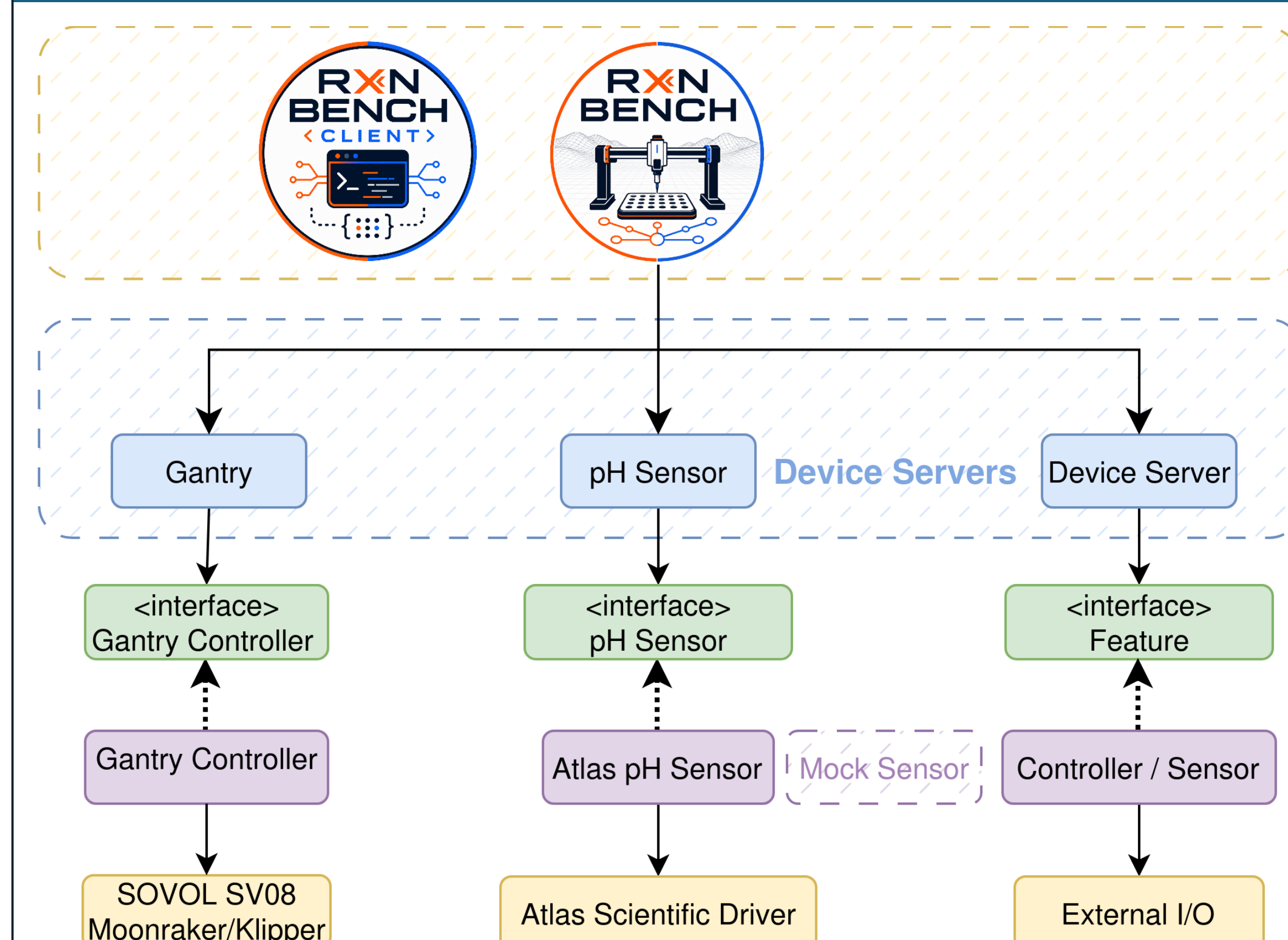
Rxn Bench is a modular and customizable laboratory automation platform designed to support programmable scientific workflows using interchangeable tools, sensors, and laboratory devices. The primary motion system is built from a repurposed **SOVOL SV08 3D printer**, combined with custom 3D-printed components and open software interfaces.

By using accessible hardware and a modular architecture, Rxn Bench aims to provide a lower-cost foundation for automated experimentation, high-throughput data collection, and the development of new laboratory automation tools via a pre-built software stack and available 3D models base for other labs to utilize, the system is intended to be expanded and adapted to a variety of laboratory environments and experimental needs.

Design Goals

Low-Cost Modular Hardware Independent Programmable Expandable

System Design



Rxn Bench utilizes a **scalable modular** architecture, which allows new devices to be added to the system smoothly.

On the Client-side Rxn Bench (*desktop application*) connects users to devices, while *Rxn Bench Client* exposes a simple API for headless control and script writing.

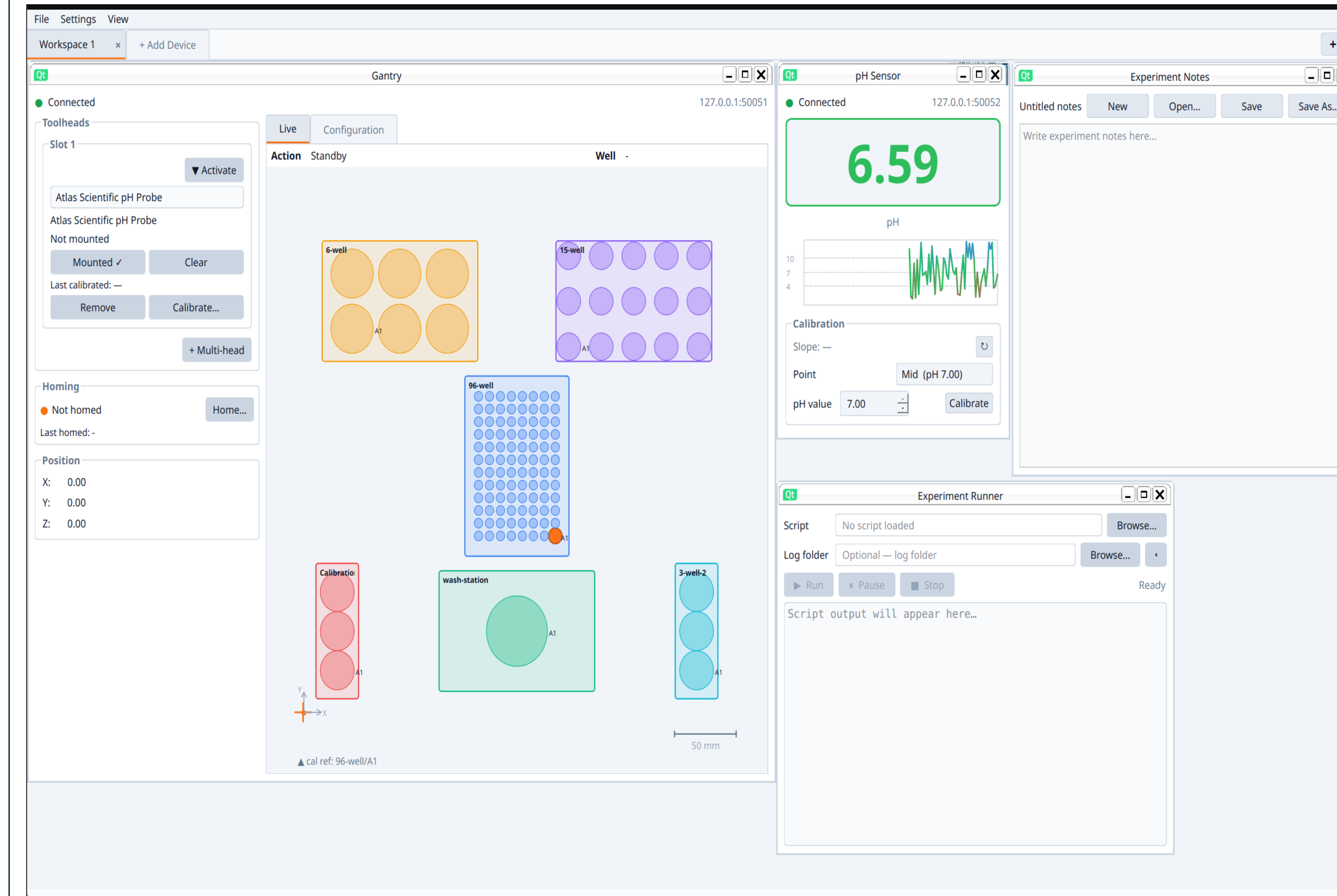
Device Servers: Each device is given its own [SiLA2](#) feature server. These servers expose commands and properties to control/communicate with devices over a network. These servers are hosted by a computer(s) in the backend, such as a Raspberry Pi

Capability Interface: Devices are integrated based on **capability** rather than specific hardware, this allows seamless integration of similar devices from different vendors or manufacturers. And the ability to create "mock" devices for remote testing/development.

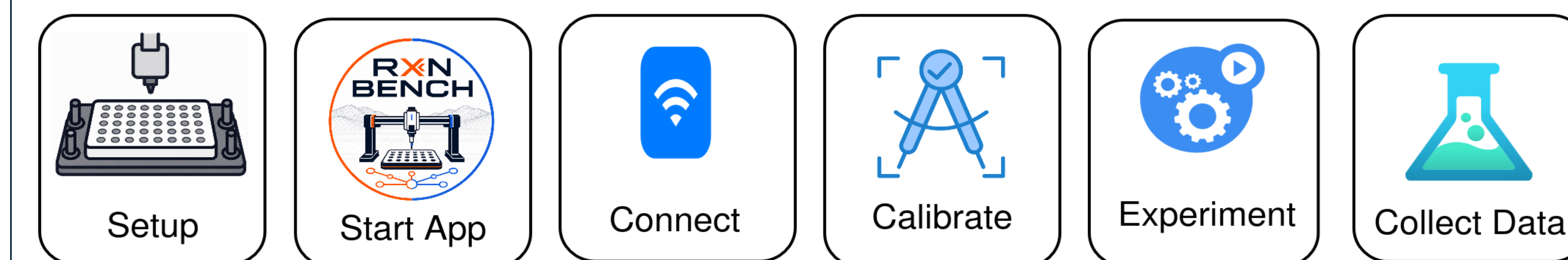
Domain Layer: Device-specific code which implements device functionality, procedures, variables, and calculations. Such as calibration, movement, etc.

Adapter Layer: Low-level adapter to establish communication with a device. Such as an instrument 'driver'.

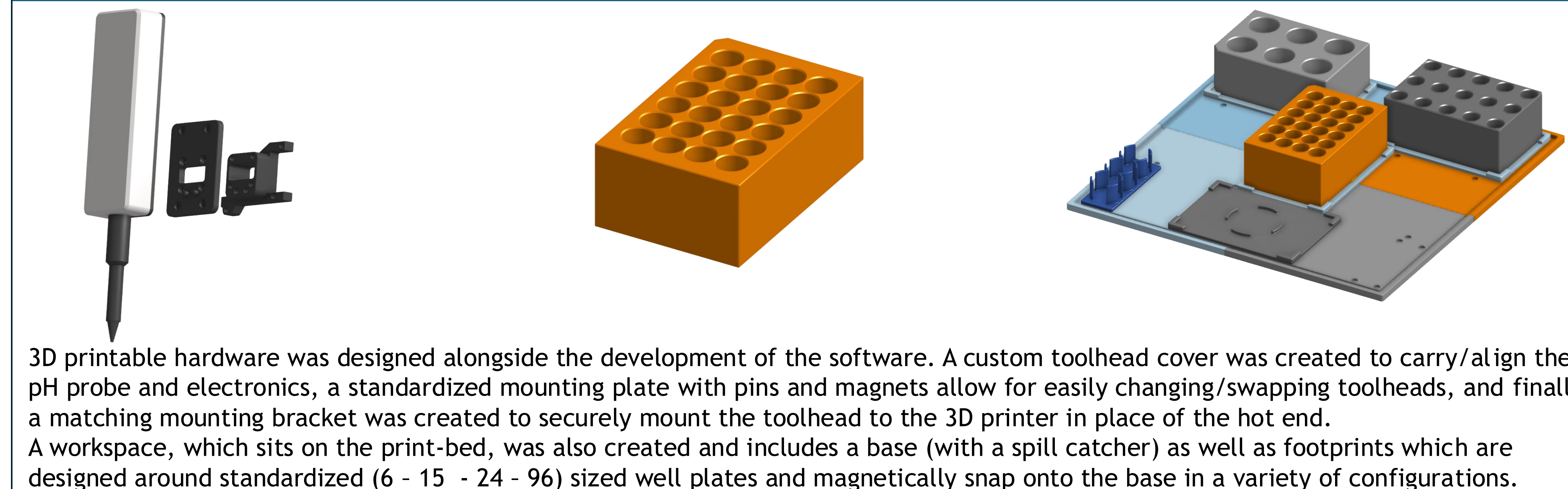
System Interface



System Workflow



Custom 3D Printed Labware Tools and Mounts



Prototype Platform



WATCH RXN BENCH IN ACTION

Scan for a short demonstration of Rxn Bench performing automatic pH calibration and measurements

<https://youtube.com/shorts/M6tpXrTsYWc>

Performance, Capabilities and Future Plans

Rxn Bench automated two- and three-point pH-probe calibration and collected measurements from more than 24 samples without requiring continuous manual operation.

Demonstrated capabilities

- Automated two- and three-point pH-probe calibration
- Automated measurement of more than 24 samples
- Configurable workspaces for multiple labware formats
- Swappable toolheads and capability-based device integration
- Programmable experiment workflows through a Python client
- Automation below \$1000.

Future development

- Multi-toolhead operation
- Automated liquid handling
- Closed-loop and self-driving experiments
- AI-assisted device integration and experiment optimization

